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Inequalities with one unknown

Solving an inequality is solving an equation with one extra rule — and an answer that is a whole interval, not a single number.

LESSON 49–50 | 2 × 40 MIN | ALGEBRA 8, §15 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Solve a linear inequality in one unknown.
- Show the solution on a number line and write it as an interval.
- Reverse the sign when multiplying or dividing by a negative number.
- Recognise an inequality with no solutions or with every number as a solution.

TEXTBOOK

National	§15, pp. 85–93
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

What a solution is

DEFINITION

A **solution** of an inequality with one unknown is any value of the unknown that makes the inequality a true statement. To **solve** the inequality is to find *all* of them.

An equation such as $2x = 6$ has one answer. The inequality $2x > 6$ has infinitely many — every number bigger than 3. So the answer is written as an **interval**, not a number.

The method

Exactly the steps you use for an equation, with one addition:

1. Clear fractions and open brackets.
2. Collect the unknowns on one side, the numbers on the other.

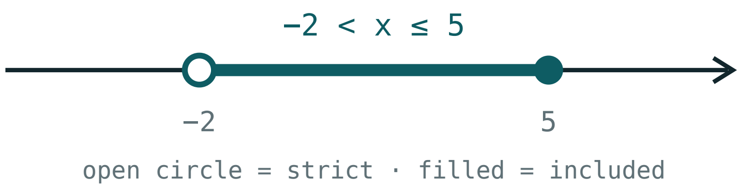
3. Divide by the coefficient of x — and if that coefficient is **negative**, **reverse the sign**.
4. Draw the answer on a number line and write the interval.

$$3x - 7 < x + 5 \Longrightarrow 2x < 12 \Longrightarrow x < 6$$

THE ONLY REAL TRAP

From $-4x \geq 12$ you get $x \leq -3$. Write the reversed sign **first**, then do the division — that way you cannot forget it.

Drawing and writing the answer



An open circle for a strict inequality, a filled circle when the endpoint is included.

INEQUALITY	ON THE LINE	INTERVAL
$x > 3$	open circle at 3, shaded right	$(3;\ +\infty)$
$x \geq 3$	filled circle at 3, shaded right	$[3;\ +\infty)$
$x < 3$	open circle at 3, shaded left	$(-\infty;\ 3)$
$x \leq 3$	filled circle at 3, shaded left	$(-\infty;\ 3]$

An infinity end is always written with a round bracket — you never reach it.

Two special answers

Sometimes the unknown disappears. Then read what is left:

$$2x + 5 > 2x + 1 \Longrightarrow 5 > 1$$

That is true whatever x is, so **every number** is a solution: the answer is $(-\infty;\ +\infty)$.

$$3x + 2 < 3x - 4 \Longrightarrow 2 < -4$$

That is false whatever x is, so the inequality has **no solutions**.

02 Worked examples

EXAMPLE 1 ***Solve $5x - 3 \le 2x + 9$***

1. $5x - 2x \le 9 + 3$

   Collect x on the left, numbers on the right.

2. $3x \le 12$

3. $x \le 4$

   Dividing by 3, a positive number — the sign stays.

4. $(-\infty; 4]$

   Filled circle at 4, shaded to the left.

ANSWER

$x \le 4$, that is $(-\infty; 4]$

EXAMPLE 2 **Solve $4 - 3x > 19$**

1. $-3x > 15$

   Subtract 4 from both sides.

2. Dividing by -3 — the sign must reverse.

   Write the new sign first.

3. $x < -5$

4. $(-\infty; -5)$

   Open circle at −5.

ANSWER

$x < -5$

EXAMPLE 3 **Solve $\frac{x-1}{2} + \frac{x}{3} \geq 1$**

1. *multiply by 6*

   The LCD of 2 and 3 — 6 is positive, so the sign stays.

2. $3(x - 1) + 2x \geq 6$

3. $3x - 3 + 2x \geq 6$

4. $5x \geq 9$, so $x \geq 1.8$

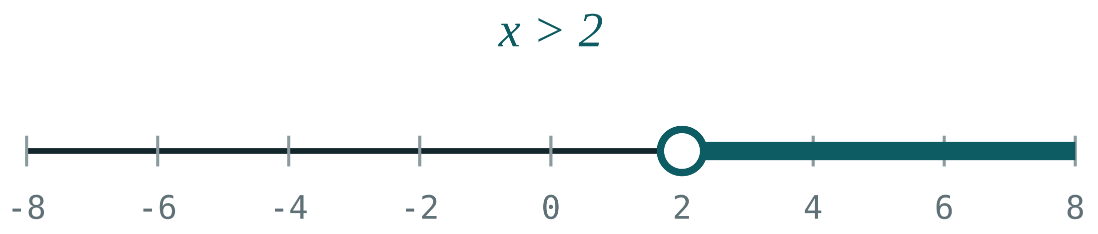
ANSWER

$x \geq 1.8$, that is $[1.8; +\infty)$

03 Interactive model

Move the boundary, flip the sign, and toggle strict — three controls cover every answer in this lesson.

Solutions on the number line



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality with one unknown	Bir noma’lumli tengsizlik	Неравенство с одним неизвестным
Solution of an inequality	Tengsizlikning yechimi	Решение неравенства
Solution set	Yechimlar to‘plami	Множество решений
Interval	Oraliq	Промежуток
Number line	Sonlar o‘qi	Числовая прямая
Equivalent inequalities	Teng kuchli tengsizliklar	Равносильные неравенства
Reverse the sign	Ishorani almashtirish	Изменить знак неравенства
No solutions	Yechimga ega emas	Нет решений
Every number	Har qanday son	Любое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$-2x > 10$ gives:

- $x > -5$
- $x < -5$
- $x > 5$
- $x < 5$

$x \geq 3$ as an interval is:

- $(3; +\infty)$
- $[3; +\infty)$
- $(-\infty; 3]$
- $[3; +\infty]$

$2x + 5 > 2x + 1$ has:

- no solutions
- one solution
- every number as a solution
- $x > 0$

$3x + 2 < 3x - 4$ has:

- no solutions
- one solution
- every number
- $x < 0$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Solve* $2x > 8$

   ANSWER $x > 4$

2. *Solve* $x + 3 \le 7$

   ANSWER $x \le 4$

3. *Solve* $x - 5 > 1$

   ANSWER $x > 6$

4. *Solve* $-x < 2$

   ANSWER $x > -2$

5. *Solve* $3x \ge -9$

   ANSWER $x \ge -3$

6. *Solve* $-2x \ge 6$

   ANSWER $x \le -3$

7. *Write* $x < 5$ *as an interval*

   ANSWER $(-\infty; 5)$

Medium · the standard question

7 PROBLEMS

1. Solve $5x - 3 \le 2x + 9$

   ANSWER $x \le 4$

2. Solve $4 - 3x > 19$

   ANSWER $x < -5$

3. Solve $2(x - 1) < x + 4$

   ANSWER $x < 6$

4. Solve $\frac{x}{3} + 2 \ge 5$

   ANSWER $x \ge 9$

5. Solve $7 - x \le 2x + 1$

   ANSWER $x \ge 2$

6. Write $x \ge -4$ as an interval

   ANSWER $[-4; +\infty)$

7. Solve $-5x + 2 < 17$

   ANSWER $x > -3$

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $\dfrac{x-1}{2} + \dfrac{x}{3} \geq 1$

   ANSWER $x \geq 1.8$

2. *Solve* $3(2x - 1) - 2(x + 4) < 5$

   ANSWER $6x - 3 - 2x - 8 < 5$, so $x < 4$

3. *Solve* $2x + 5 > 2(x + 1)$

   ANSWER Every number — $5 > 2$ is always true.

4. *Solve* $3(x - 2) < 3x - 4$

   ANSWER $3x - 6 < 3x - 4$ reduces to $-6 < -4$, which is always true — every number is a solution.

5. *Find the smallest whole number satisfying* $4x - 7 > 9$

   ANSWER $x > 4$, so $x = 5$

6. *Find the largest whole number satisfying* $5 - 2x \geq -3$

   ANSWER $x \leq 4$, so $x = 4$

7. *For which* x *is* $\dfrac{2x-1}{3} - \dfrac{x}{2} \leq 0$?

   ANSWER Multiply by 6: $4x - 2 - 3x \leq 0$, so $x \leq 2$.

07 Homework

Homework — 6 problems

Algebra 8, §15, pp. 85–93. Draw the number line for every answer.

1. *Solve* $4x - 5 \geq x + 7$
2. *Solve* $6 - 5x > 21$
3. *Solve* $3(x + 2) \leq 2x + 10$

4. *Solve* $\frac{x}{4} - 1 < 2$

5. *Find the smallest whole number with* $7x - 3 > 25$

6. *Solve* $2(x - 3) > 2x - 5$ *and explain your answer*